THE SPECTRAL STRUCTURE OF FRACTIONAL FIELDS

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Abstract. For primitive-root primes ($\mathrm{ord}_p(b) = p-1$), we prove that the cross-alignment matrix, reordered by discrete-logarithm offset, is a circulant whose eigenvalues are the Fourier transform of the digit-equality autocorrelation. For primes with exactly two cosets and zero within-coset autocorrelation at nonzero lags, the spectrum has two eigenvalue levels determined by the cross-coset match count. The multi-coset case is described observationally.

1. Introduction

The cross-alignment matrix $\mathbf{A}(p)$ introduced in [1] is a symmetric, positive semidefinite matrix on the fractions of a prime p in base b, whose (i, j) entry is the digit-equality proportion between i/p and j/p. For digit-partitioning primes ($p \leq b+1$), the matrix is the identity, every fraction is a stranger to every other, and the eigenvalue spectrum is trivially flat. For non-digit-partitioning primes, the matrix is no longer the identity. Its eigenvalue spectrum encodes the interference geometry of the field, but the spectrum was not given a closed form in [1].

This paper identifies the eigenvalues. The main result (Theorem 2) is that for primes p where b is a primitive root modulo p, the matrix $\mathbf{A}(p)$, after a discrete-logarithm reordering of its rows and columns, becomes a circulant matrix whose first row is the cyclic autocorrelation of the base repetend $1/p$, divided by the cycle length $L = p - 1$. Standard properties of circulants [2] then identify the eigenvalues as the discrete Fourier transform of the first row. The eigenvalue spectrum of $\mathbf{A}(p)$ is a character expansion of the repetend's autocorrelation.

For primes p where b is not a primitive root, the fractions partition into $C = (p-1)/L$ orbits of $\langle b \rangle$, and the matrix decomposes into within-orbit and between-orbit blocks. We treat the two-orbit case in detail (Theorem 4) and show that under the natural sufficient conditions, which are satisfied at $p = 13$ in base 10, the spectrum collapses to two

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eigenvalue levels determined by an involution. The general multi-orbit case is described observationally in §5.

The connection to harmonic analysis on finite groups is explicit: the eigenvalues are inner products of the autocorrelation function with the additive characters of $\mathbb{Z}/L\mathbb{Z}$. This identifies the eigenvalue spectrum as a Fourier transform on a finite cyclic group, which is the natural starting point for the harmonic analysis pursued in the next several papers.

2. The Circulant Theorem

Definition 1. The *cyclic autocorrelation* of the base repetend $(d_0, d_1, \ldots, d_{L-1})$ of $1/p$ is the sequence

$$R(\ell) = |\{j \in \{0, \ldots, L-1\} : d_j = d_{(j+\ell) \bmod L}\}|, \qquad \ell = 0, 1, \ldots, L-1.$$

Note that $R(0) = L$ (every position matches itself) and $R(\ell) \geq 0$ for all ℓ.

Theorem 2. *Let p be a prime for which b is a primitive root ($\mathrm{ord}_p(b) = p-1$). Then the cross-alignment matrix $\mathbf{A}(p)$, reordered by discrete-logarithm offset, is a circulant matrix with first row*

$$\left(\frac{R(0)}{L}, \frac{R(1)}{L}, \frac{R(2)}{L}, \ldots, \frac{R(L-1)}{L}\right).$$

Its eigenvalues are given by the discrete Fourier transform [2]*:*

$$\lambda_j = \frac{1}{L}\sum_{\ell=0}^{L-1} R(\ell)\,\omega^{j\ell}, \qquad j = 0, 1, \ldots, L-1,$$

where $\omega = e^{2\pi i/L}$.

Proof. When b is a primitive root modulo p, every fraction k/p is a cyclic rotation of $1/p$ in the sense that the repetend of k/p in base b is the repetend of $1/p$ shifted by $\log_b k$ positions modulo L (where $\log_b$ denotes the discrete logarithm with respect to b modulo p). Ordered by rotation offset, the fractions are $1/p,\ b/p,\ b^2/p,\ \ldots,\ b^{L-1}/p$ (each reduced modulo p).

In this ordering, the (i, j) entry of $\mathbf{A}(p)$ is the digit-equality proportion between b^i/p and b^j/p, which is the proportion of positions in the repetend of $1/p$ where the digit at position ℓ agrees with the digit at position $\ell+(j-i) \bmod L$. By the definition of R, this proportion is $R(j-i \bmod L)/L$. The (i, j) entry depends only on the difference $j-i$ modulo L, so the matrix is circulant with first row $(R(0)/L,\ R(1)/L,\ \ldots,\ R(L-1)/L)$.

The eigenvalues of a circulant are the discrete Fourier transform of its first row [2], which yields the stated formula. □

Corollary 3. *If p is digit-partitioning ($p \leq b+1$), every eigenvalue of $\mathbf{A}(p)$ equals 1.*

Proof. By [1], $\mathbf{A}(p) = I_{p-1}$ for digit-partitioning p, so the spectrum is the constant sequence 1. For primitive-root digit-partitioning primes, the same conclusion follows from Theorem 2: the digit-partitioning condition forces $R(\ell) = 0$ for $\ell \neq 0$ (no two distinct positions of the repetend share a digit), and the DFT of $(L, 0, 0, \ldots, 0)/L$ is the constant sequence 1. □

3. Worked Example: $p = 7$ in Base 10

In base 10, the prime $p = 7$ is digit-partitioning ($p \leq b+1 = 11$) and primitive-root ($\mathrm{ord}_7(10) = 6$). The repetend of $1/7$ is 142857, of length $L = 6$. By Theorem 2, $\mathbf{A}(7)$ in the discrete-logarithm ordering is a 6×6 circulant whose first row is $(R(0)/6, R(1)/6, \ldots, R(5)/6)$.

The autocorrelation values are computed by direct comparison of 142857 with its rotations. At lag $\ell = 1$ the rotated sequence is 428571, and the position-wise comparison (1:4, 4:2, 2:8, 8:5, 5:7, 7:1) yields zero matches. The same calculation at every lag $\ell \in \{1, 2, 3, 4, 5\}$ gives zero matches. Hence $R(0) = 6$ and $R(\ell) = 0$ for $\ell \neq 0$, and the first row of the circulant is $(1, 0, 0, 0, 0, 0)$.

The DFT of this row is the constant sequence $(1, 1, 1, 1, 1, 1)$. All six eigenvalues of $\mathbf{A}(7)$ equal 1, in agreement with the digit-partitioning case of Corollary 3.

4. The Two-Orbit Case

When $\mathrm{ord}_p(b) = L < p-1$, the fractions partition into $C = (p-1)/L$ orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$. For $C = 2$, the cross-alignment matrix has additional structure that we make precise below.

Theorem 4. *Let p be a prime with exactly two orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$, each of size $L = (p-1)/2$. Suppose*

1. *the within-orbit autocorrelation $R_C(\ell)$ (the digit-equality count between rotations within the same orbit) satisfies $R_C(\ell) = 0$ for $\ell \neq 0$, and*
2. *each fraction in one orbit matches exactly one fraction in the other orbit at exactly m positions, with all other cross-orbit pairs matching at zero positions.*

Then the eigenvalues of $\mathbf{A}(p)$ *are*

$$1 + \frac{m}{L} \quad \textit{and} \quad 1 - \frac{m}{L},$$

each with multiplicity $L = (p-1)/2$.

Proof. Under these conditions, every diagonal entry of $\mathbf{A}$ is 1, every within-orbit off-diagonal entry is 0 (from condition (1)), and every cross-orbit off-diagonal entry is either m/L (for the unique partner) or 0 (otherwise). We can therefore write

$$\mathbf{A}(p) = \mathbf{I} + \frac{m}{L}\,\mathbf{P},$$

where $\mathbf{P}$ is the permutation matrix that pairs each fraction with its unique cross-orbit partner. Since the pairing is symmetric, $\mathbf{P}$ is an involution ($\mathbf{P}^2 = \mathbf{I}$), so its eigenvalues are ± 1, each with multiplicity equal to half its dimension, i.e., multiplicity L each. The eigenvalues of $\mathbf{A}$ follow as $1 \pm m/L$. □

Proposition 5. *The hypotheses of Theorem 4 are satisfied at* $p = 13$ *in base* 10*, with* $m = 2$ *and* $L = 6$*. The eigenvalues of* $\mathbf{A}(13)$ *are* $4/3$ *and* $2/3$*, each with multiplicity* 6*.*

Proof. The order of 10 modulo 13 is 6, so $L = 6$ and $C = 2$. The two orbits of $\langle 10 \rangle$ in $(\mathbb{Z}/13\mathbb{Z})^*$ are $\{1, 10, 9, 12, 3, 4\}$ and $\{2, 7, 5, 11, 6, 8\}$. The repetends within the first orbit are the rotations of 076923, and within the second orbit they are the rotations of 153846.

Within-orbit comparison: comparing 076923 with each of its rotations 769230, 692307, 923076, 230769, 307692 position by position gives zero matches in every case. Hence $R_C(\ell) = 0$ for $\ell \neq 0$, verifying condition (1).

Cross-orbit comparison: comparing $1/13 = 076923$ with each member of the second orbit gives the following position-wise match counts:

partner	matches
$2/13 = 153846$	0
$5/13 = 384615$	0
$6/13 = 461538$	0
$7/13 = 538461$	0
$8/13 = 615384$	0
$11/13 = 846153$	2

The unique partner of $1/13$ in the other orbit is $11/13$, matching at exactly 2 positions (positions 3 and 6). The same pattern holds for every other fraction in the first orbit, by replaying the calculation: each

fraction has a unique cross-orbit partner matching at exactly 2 positions and matches every other cross-orbit fraction at 0 positions. This verifies condition (2) with $m = 2$.

By Theorem 4, the eigenvalues of $\mathbf{A}(13)$ are $1 + 2/6 = 4/3$ and $1 - 2/6 = 2/3$, each with multiplicity 6. □

Whether the hypotheses of Theorem 4 hold for every two-orbit prime, or only for a subfamily that includes $p = 13$, is not determined here.

5. The General Multi-Orbit Case

For primes with $C = (p-1)/L > 2$ orbits, the cross-alignment matrix has a block decomposition that we describe but do not prove in full generality.

Proposition 6. *For any prime p with C orbits of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$, each of size L, the cross-alignment matrix admits the decomposition*

$$\mathbf{A}(p) = \mathbf{W}(p) + \mathbf{X}(p),$$

where $\mathbf{W}(p)$ is the block-diagonal within-orbit contribution (a direct sum of C circulant $L \times L$ matrices, one per orbit, each governed by the within-orbit autocorrelation) and $\mathbf{X}(p)$ is the block-off-diagonal between-orbit contribution (whose (i, j) block records the cross-orbit alignments between orbit i and orbit j).

Proof. The block partition follows from grouping the rows and columns of $\mathbf{A}(p)$ by orbit. Within each orbit of size L, the rotation argument of Theorem 2 applies verbatim, since fractions within the same orbit are rotations of one another, so each diagonal block is a circulant whose first row is $R_C(\ell)/L$ for that orbit. The cross-orbit blocks contain the digit-equality proportions between fractions of different orbits and have no cyclic structure in general. □

The within-orbit eigenvalues are the DFT of $R_C(\ell)/L$ for each orbit, by Theorem 2 applied within the orbit. The cross-orbit blocks add a perturbation whose eigenvalues depend on the joint structure of all C orbits.

Computation for all primes $p \leq 37$ in base 10 shows that the spectrum of $\mathbf{A}(p)$ is determined by the within-orbit autocorrelations $R_C(\ell)$ together with the cross-orbit alignment data, in the sense that two primes with the same orbit structure and the same alignment data have the same spectrum. A closed-form expression for the multi-orbit eigenvalues in terms of these data is not given here.

6. Connection to Harmonic Analysis

The DFT that produces the eigenvalues is a sum over characters of $\mathbb{Z}/L\mathbb{Z}$:

$$\lambda_j = \frac{1}{L}\sum_{\ell=0}^{L-1} R(\ell)\,\chi_j(\ell),$$

where $\chi_j(\ell) = e^{2\pi i j\ell/L}$ is the j-th character. The eigenvalue λ_j is therefore the inner product of the autocorrelation function with the character χ_j.

This identifies the eigenvalue spectrum of the cross-alignment matrix as a *character expansion* of the repetend's autocorrelation, in the sense of harmonic analysis on finite abelian groups [3]. The digit-partitioning property corresponds to a flat character expansion (all coefficients equal). Non-DP primes exhibit character-dependent structure.

For primitive-root primes, where $L = p - 1$, the additive characters of $\mathbb{Z}/L\mathbb{Z}$ used here are not the Dirichlet characters of $(\mathbb{Z}/p\mathbb{Z})^*$, but the two character groups are isomorphic via the discrete logarithm: the map $\log_b : (\mathbb{Z}/p\mathbb{Z})^* \to \mathbb{Z}/L\mathbb{Z}$ intertwines a multiplicative character of $(\mathbb{Z}/p\mathbb{Z})^*$ with an additive character of $\mathbb{Z}/L\mathbb{Z}$. The eigenvalues of $\mathbf{A}(p)$ can therefore be re-expressed as evaluations of Dirichlet characters modulo p on the autocorrelation pulled back through the discrete logarithm. The Dirichlet character framework [4] provides the analytic machinery for relating these character sums to L-functions, and is the natural setting in which the next several papers in the chain are developed.

7. Remarks

The autocorrelation in the primitive-root case. For primitive-root primes, the cyclic autocorrelation $R(\ell)$ determines the full eigenvalue spectrum (Theorem 2). For multi-coset primes, the within-coset autocorrelation alone does not suffice: the between-coset cross-correlation data is also needed (as seen at $p = 13$ in base 10, where $R(\ell) = 0$ for $\ell \neq 0$ yet the spectrum is not flat).

From spectra to L-functions. The expression $\lambda_j = L^{-1}\sum R(\ell)\chi_j(\ell)$ resembles a Gauss-type sum. If the autocorrelation $R(\ell)$ can be expressed in terms of the Legendre symbol or other multiplicative characters, the eigenvalues become evaluations of Dirichlet L-functions at specific points. Whether this connection yields non-trivial analytic information about the fractional field is open.

References

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